Requirements for 70Z03826IB0000007
Attachment 1

REQUIRED CAPABILITIES

3.1 Structural and Fire Safety Compliance:

1. Demonstrated compliance with NFPA 33, including the use of non-combustible or documented fire-retardant materials for the enclosure.
2. System's ability to be securely anchored or stabilized in a typical aircraft hangar environment.
3. Compliance with NFPA 409 and NFPA 410 regarding operations within an aircraft hangar.

3.2 Ventilation and Environmental Controls:

1. Capability to maintain a consistent negative pressure environment in the work area of no less than -0.10 inches water gauge (in WG) static pressure with respect to the surrounding area during sanding operations, and -0.05 in WG when not sanding. Capability to maintain a consistent negative pressure environment in the transition/decontamination (doffing) area of no less than -0.05 in WG relative to the clean area.
2. Exhaust system design, including ducting and fan construction (e.g., explosion-proof per NFPA 91).
3. Filtration system capabilities, specifically addressing control of hexavalent chromium and cadmium-contaminated dust. Detail the filter stages available (e.g., pre-filter, HEPA, etc.) and the inclusion of a manometer or other device for monitoring pressure differentials to monitor filter life.
4. Integration with source-capture vacuum systems.

3.3 Electrical and Ignition Source Control:

1. Compliance with NFPA 70 (NEC) for electrical components, specifically Class I, Division 1 or 2 ratings for all equipment within the hazardous location.
2. Description of lighting solutions (e.g., explosion-proof fixtures, external lighting through sealed panels).

3.4 Safety and Operational Features:

1. Information on integrated fire suppression systems (automatic or manual).
2. Description of a multi-chamber design (e.g., clean entry, transition/decontamination, and work zones) to control contamination.
3. Availability of intrinsically safe communication systems for personnel.
4. Details on training, maintenance procedures, and recommended inspection schedules for the system.

3.5 Additional requirements

1. Three chambered system – clean area, transition/decontamination area, and work area.

2. Dedicated ports or pass-throughs for connecting HEPA-filtered industrial vacuums for source capture of sanding dust directly from the tool.
3. Dedicated ports or pass-throughs for connecting downdraft tables to power for source capture of sanding dust directly from the tool.
4. HEPA filtration 99.97% capture rate at 0.3 microns.
5. Negative pressure as described in section 3.2.1.
6. Temperature and humidity control.
7. Space for 4-6 downdraft tables with 3ft clearance on all sides. Tables are approximately 3'x6' each.
8. Interior surfaces shall be smooth, designed and installed to prevent pockets that can trap residues, and designed to facilitate ventilation and cleaning.

<u>References to consider</u>:

1. NFPA 409: Standard on Aircraft Hangars, 2022 Edition - Chapter 12 Paint Hangars
2. NFPA 33: Standard for Spray Application Using Flammable or Combustible Materials, 2024 Edition – Chapter 14.4 Inflatable Finishing Workstations and Chapter 18 Spray Application Operations in Membrane Enclosures
3. NFPA 410: Standard on Aircraft Maintenance, 2025 Edition – Chapter 7 Aircraft External Cleaning, Painting, and Paint Removal, Chapter 10 Fire Protection
4. NFPA 91: Standard for Exhaust Systems for Air Conveying Vapors, Gases, Mists, and Particulate Solids, 2026 Edition
5. 29 CFR 1910.94 - Ventilation, Table G-10
6. 29 CFR 1910.107 - Spray finishing using flammable and combustible materials
7. Aeronautical Engineering Safe Work Practices Guide, CGTO PG-85-00-2210-A
8. Aviation Corrosion Control Facilities (CCF) (Rev A), SILC-CSTO-36-71 91 11 13-07
9. Aircraft Corrosion Control and Paint Facilities (2012), UFC 4-211-02
10. EPA/NESHAP 6H